

Module	Water Infrastructure: Channel Design
Topic	Shear Stress Calculations

Shear stress of flow

Shear stress is represented by the τ (tau) symbol and represents the shear stress caused by the flow of water. Figure 1 is used to describe the forces acting on the bed of a channel.

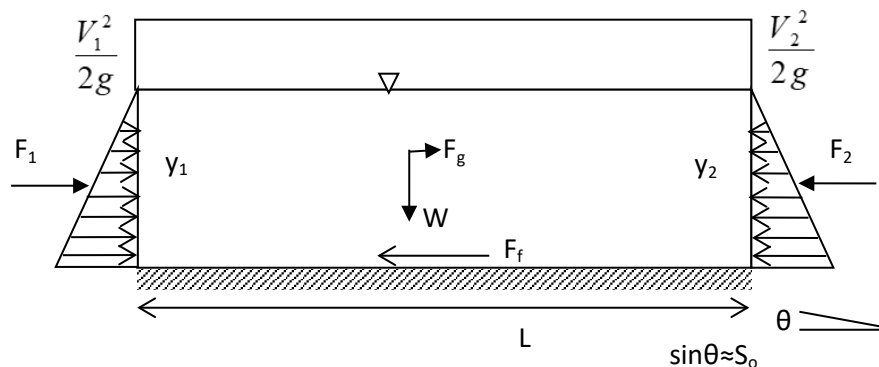


Figure 1. Forces of water on the bed of a channel

The force of friction and gravity are denoted F_f and F_g respectively and can be described in a control volume as:

$$F_f = -\tau_o PL \quad (\text{negative sign represent force acting opposite to direction of flow})$$

$$F_g = \gamma AL \sin \theta$$

Where,

P = perimeter of a channel

L = length

A = area

γ = specific weight of water ($\sim 9810 \text{ N/m}^3$)

Note: The weight of fluid in a control volume is $(\gamma AL) = W$

For a steady uniform flow the general form of the integral momentum equation can be simplified to:

$$F_1 + F_g + F_f - F_2 = 0$$

For uniform flow: $F_1 = F_2$

Which simplifies to $F_g + F_f = 0$

$$\gamma AL \sin \theta - \tau_o PL = 0$$

for small θ (such as in most rivers/channels), $\sin \theta \approx S_o$

$$\gamma ALS_o = \tau_o PL$$

$$R=A/P$$

$\tau_o = \gamma R S_o$ note: This is an important equation to remember.

Shear velocity can be further defined as the following:

$$u_o^* = \sqrt{\frac{\tau_o}{\rho}} = \sqrt{g R S_o}$$

Shear stress and Manning's

Shear stress for turbulent flow can also be expressed as a function of density, velocity, and a resistance coefficient (C_f).

$$\tau_o = C_f \rho \left(\frac{V^2}{2} \right)$$

$$C_f \rho \left(\frac{V^2}{2} \right) = \gamma R S_o$$

Solving for velocity gives:

$$V = \sqrt{\frac{2g}{C_f}} \sqrt{R S_o}$$

this is a form of the Chezy equation:

$$V = C \sqrt{R S_o}$$

Where C is the Chezy coef.

Manning's derived an empirical relationship for C based on experiments:

$$C = \frac{1}{n} R^{1/6}$$

so we have Manning's formula:

$$V = \frac{1}{n} R^{2/3} S^{1/2}$$

Threshold conditions

Threshold conditions refer to the flow conditions (discharge, velocity, or depth) that will initiate sediment motion of grain forming the boundary layer of the channel. This is important in the design of a stable channel so no scouring will occur or so that no deposition will occur. Sometime both conditions are required. For example in an irrigation channel coarse material will be used to protect the banks and bed and at the same time the channel will be designed so no deposition occurs.

Forces on a grain:

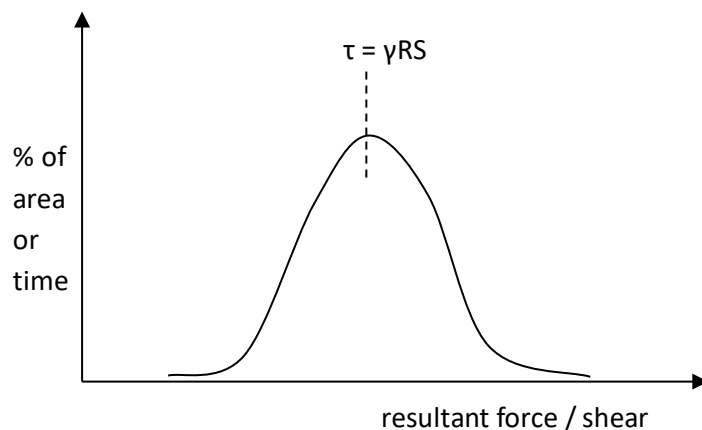
Forces on a grain arise from drag and lift exerted on the grain from the fluid.

Drag – from skin friction (small component) and from pressure difference arising from flow separation.

Lift – from pressure differences between stagnation pressures in the bed and pressure due to flow and curved streamlines over the grain. Contributions also occur from vertical velocities in turbulent flow.

Turbulent flow: will give rise to fluctuations in both lift and drag with time. Point of application and direction of forces will also vary with time.

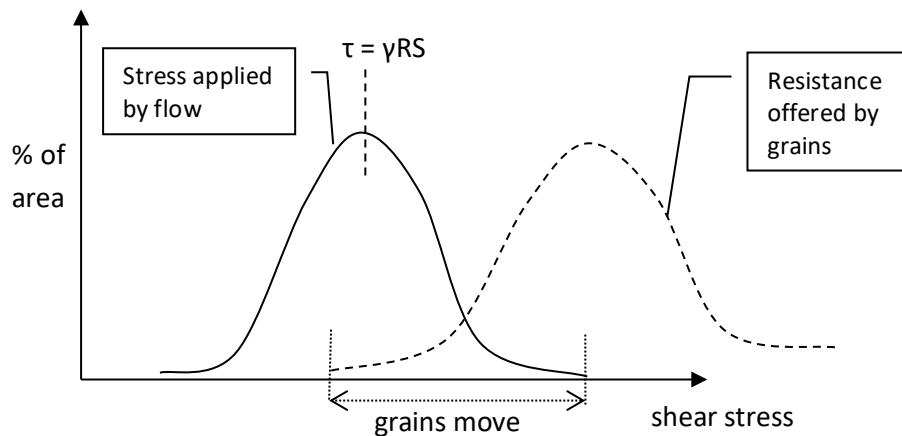
Particle shape, size, location, and orientation will also affect the forces on particular particles. Thus a distribution of forces in both time and space would look as shown below:



Resisting forces: comprise of the weight and any interference forces (mechanical or cohesive) between grains.

Cohesion is a very difficult subject to cover since it depends on water and chemical properties of particles. For this reason we are mostly confined to non-cohesive particles (sands, gravels) and we thus exclude clay sized particles.

Combining the applied and resisting forces for a given flow and bed of particles we obtain the following diagram:



If the diagrams overlap there will be motion of grains! If they do not overlap, then there is no motion. When the distributions of grains have long “tails”, the difficulty is in deciding on a true critical condition. It is most likely that regardless of the design, something will move.

Critical conditions: We can therefore define critical conditions as those for which an appreciable number of grains are in motion. “Appreciable” will depend on the particular application and will vary according to the amount of motion one can tolerate. This will depend on what you are designing.

Studies on threshold or entrainment of particles

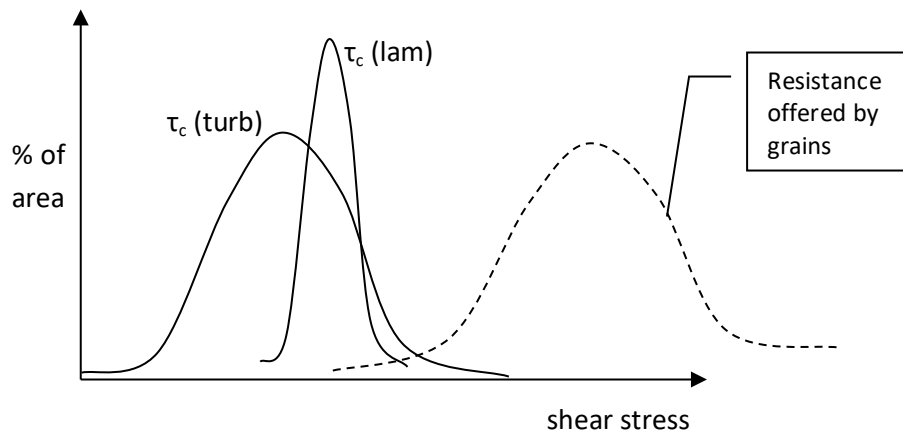
Various studies have been carried out on the problem of the threshold or entrainment of particles. Two fundamental studies are those conducted by White and Shields.

White: White considered a particular grain and assumed it became unstable and moved by rotating about a downstream point of contact. Threshold conditions thus implied equality between resisting moment and overturning moment.

He came up with the following relationship where τ_c indicates critical or threshold conditions:

$$\tau_c = k(\gamma_s - \gamma)d \tan \phi$$

1. ϕ on average over the whole bed is equal to the angle of repose. When the bed is tilted to the angle of repose, the grains move at zero shear because the angle shown as ϕ has gone to zero.
2. White's results shows $\tau_c \propto d$. Submerged weight (W) $W \propto d^3$ and with $\tau_c \propto u_c^2$ then, $u_c^2 \propto W^{1/3}$ and $u_c^2 \propto W^{1/6}$
3. Experiments with laminar flow showed White that $k=0.18$ and for turbulent flow because of the width of the frequency plot k was only 50 to 60% of the laminar value. So in turbulent flow average threshold shear stresses are smaller. Measurements in turbulent flows show that shear stresses can reach 2 -2.5 times the mean value.



So, as a result some grain motion will occur at lower values of mean shear although the shear stresses causing movement are the same in laminar and in turbulent flows. White considered the true value to be:

$$\tau_c = 0.18(\gamma_s - \gamma)d \tan \phi$$

or

$$\frac{\tau_c}{(\gamma_s - \gamma)d} = 0.18 \tan \phi$$

The dimensionless number $\frac{\tau_c}{(\gamma_s - \gamma)d}$ is known as the entrainment function and plays a very significant role in sediment transport.

Shields diagram

The Shields diagram (Shields 1936) was developed from experimental measurements to predict the entrainment (incipient motion) of particles due to flow. The Shields diagram shows the functional relationship between the Boundary Reynolds number and the entrainment function. Shields (Shields 1936) argued that sediment grains would move when the drag force was equal to the weight resistance. This is a sliding rather than rolling failure.

Limitations of Shields work are the following:

- only flat beds were considered
- threshold conditions were established by extrapolating sediment transport rate measurements back to zero
- uniform grain sizes were used in all experiments
- only fully developed turbulent flows were investigated

In practice these conditions are not always met. A more serious limitation is that Shields always had R/d very large. His results do not work for R/d less than 5 or for very steep slopes $S > 0.01$

However, it is the best we have for sand size particles, so use Shields!

Shields Entrainment function:

$$\frac{1}{\Psi} = \frac{\tau_o}{(\gamma_s - \gamma)d} = \frac{RS}{(S_s - 1)d} = \frac{u_*^2}{(S_s - 1)gd} = \frac{\tau_o d^2}{(\gamma_s - \gamma)d^3}$$

The final variation of the equation shows the parameter to be the ratio of the drag force to the immersed weight. It is thus a measure of mobility of a particle in a given flow and thus an important parameter. It is sometimes referred to as the mobility number.

Boundary Reynolds number:

$$\frac{u_* d}{\nu}$$

The $\frac{u_* d}{\nu}$ is a grain Reynolds number based on shear velocity.

For $Re_* > 400$:

$$\frac{\tau_c}{(\gamma_s - \gamma)d} = 0.056 \quad (\text{see Shields diagram in Figure 2})$$

with a viscosity of $\nu = 10^{-6} \text{ m}^2/\text{s}$ you obtain $d > 6\text{mm}$. So, for coarse material:

$$\text{Shields: } \tau_c = 0.056(\gamma_s - \gamma)d$$

$$\text{White: } \tau_c = 0.18(\gamma_s - \gamma)d \tan \phi \quad \text{and with } \phi = 35^\circ \quad \tau_c = 0.126(\gamma_s - \gamma)d$$

This makes sense because if you recall White was for laminar flow and Shields for turbulent flow so the ratio of about 2:1 in coefficients is about right. Further, White equated moments and Shields equated forces, each having his own idea of the type of initial movement. The agreement in results suggests that neither describe the physics very well and the experimental constants take up the slack.

Coarse Material (d>6mm)

$$\text{Since } \tau_o = \gamma RS_o \text{ and } \tau_o = 0.056(\gamma_s - \gamma)d$$

$$d = \frac{RS}{(S_s - 1)0.056}$$

$$d = 11RS \quad (S_s = 2.65)$$

You must use a safety factor of 1.2 or 1.25.

Finer material

Use the value of:

$$\frac{d}{\nu} \sqrt{0.1 \left(\frac{\gamma_s}{\gamma} - 1 \right) g d}$$

established by the ASCE Sedimentation Manual (1975) and show in Figure 2, where d is in metres.

otherwise you can use USBR curve with

$$\tau_{(N/m^2)} = 0.755 d_{75(mm)}^{1/6}$$

or

d = 12.7 RS (this includes a safety factor of 1.15)

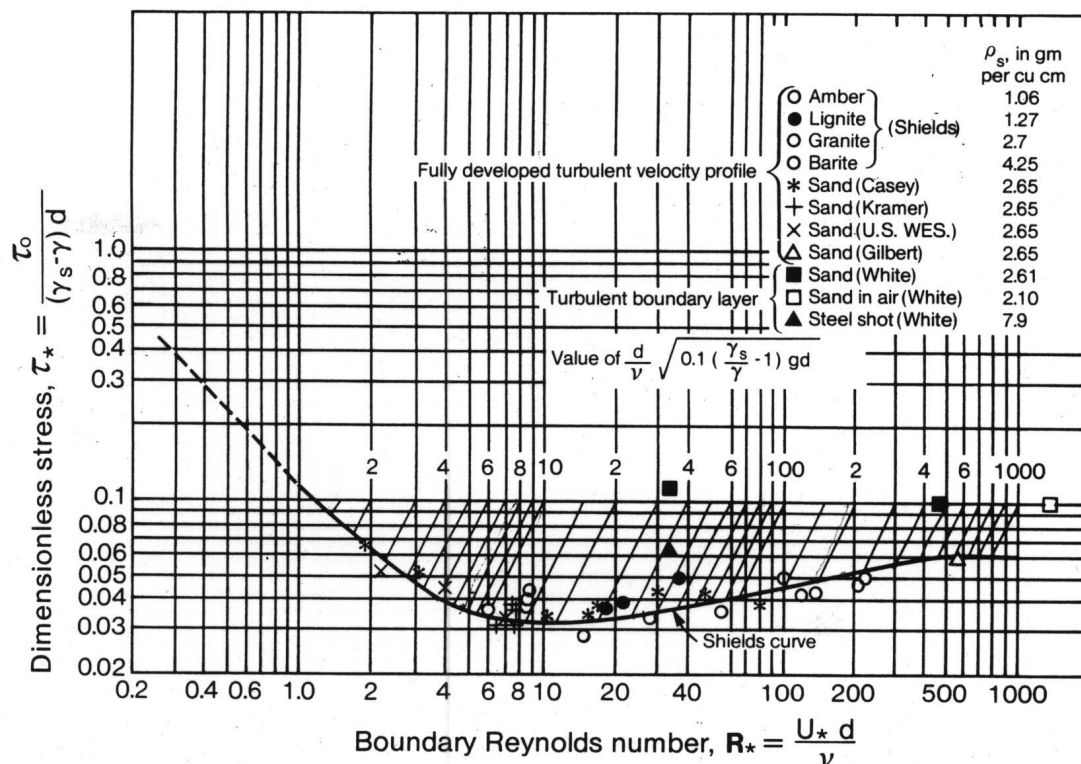


Figure 2. Shields diagram (Shields 1936) as illustrated in (Chang 1988).

Roughness and Stricklers formula

Channel design requires an estimate of roughness. For stable channels of non-cohesive sand sized and larger without bed forms use Strickler's formula:

$$n = 0.012 d_{75(mm)}^{1/6}$$

or

$$n = 0.038 d_{75(m)}^{1/6}$$

To include bed form roughness such as riffles, islands, bends, etc. as occur in gravel bed rivers, the following formula is preferred:

$$n = 0.059 d_{50(mm)}^{1/6}$$

Example:

As a mitigation measure for a logging resource consent you are required to provide a channel to convey 0.1 (m³/s)/m of clear water for a fish hatchery. The preferred line for the channel will have a slope of 0.0002. Recommend a sediment size to line the channel.

Note: For this problem you will need to use Manning's, Strickler, and the Shields diagram.

Manning Equation: $q = \frac{1}{n} R^{5/3} S^{1/2}$

Because bed must be stable we will assume $n = 0.012 \cdot d_{75(mm)}^{1/6}$ (Strickler)

Hence,

$$q = \frac{1}{0.012 \cdot d_{75(mm)}^{1/6}} R^{5/3} S^{1/2}$$

$$R = \left[\frac{0.012 \cdot d_{75(mm)}^{1/6} \cdot 0.1}{(0.0002)^{1/2}} \right]^{3/5}$$

$$R = 0.2276 \cdot d_{75(mm)}^{1/10}$$

$$\therefore RS = 4.55 \times 10^{-5} \cdot d_{75(mm)}^{1/10}$$

$$\text{and } \tau = \gamma RS, \text{ thus } \tau = \gamma \cdot 4.55 \times 10^{-5} \cdot d_{75(mm)}^{1/10}$$

where $\gamma = 9810 \text{ N/m}^3$ (specific weight at 20 degrees)

Use Figure 4 to plot where 1 mm and 10 mm particles would lie on the graph (1 mm and 10 mm were chosen randomly to have one value on top of the entrainment lines and one below the lines).

for $d = 1 \text{ mm}$:

$$\tau = 0.446 \text{ N/m}^2$$

for $d = 10 \text{ mm}$

$$\tau = 0.561 \text{ N/m}^2$$

Draw a line between them and see where it intersects the Shields curve (or relevant entrainment line) on the diagram. The diameter of the particle at the point of intersection with the entrainment line should be around:

$$d_{75(mm)} = 6 \text{ mm}$$

Then allow for a factor of safety. The actual $d_{75(mm)}$ value will depend on what material is available.

Finite width Channels

The shear stress varies around the perimeter of the channel and thus introduces a complexity into the design. For trapezoidal channels the shear stresses can be represented as in Figure 3.

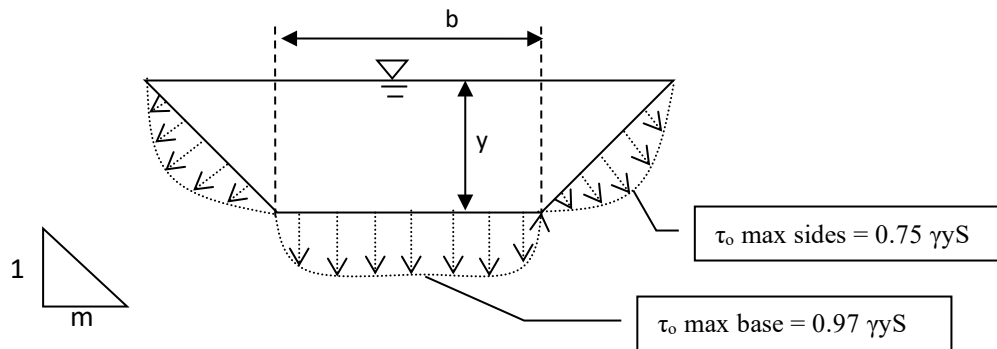


Figure 3. Estimated flow shear stress on bottom and sides of a trapezoidal channel.

Some texts give different coefficients as functions of b/y ; 0.75 and 1.0 are recommended for $m < 2.5$. For flatter sides the maximum stress tends to increase.

Critical stresses

On the base of the channel we can use a Shields value:

$$\tau_{cb} = 0.056(\gamma_s - \gamma)d \quad (\text{for coarse material, } d > 6\text{mm})$$

On the sides of the channel the gravitational component causes a reduction in resistance so that

$$\tau_{cs} = \left[1 - \frac{\sin^2 \alpha}{\sin^2 \phi} \right]^{1/2} \cdot \tau_{cb}$$

or

$$\tau_{cs} = k \tau_{cb} \quad (k < 1)$$

Where k is the tractive force ratio, α the side slope angle $= \tan^{-1}(1/m)$ and ϕ is the angle of repose (generally 35°).

This poses the question of whether the sides will become unstable first or the base as the discharge increases. This can be determined by the following:

- Comparison of shear stresses from sides to base: $\tau_{os} = 0.75 \tau_{ob}$
- Comparison of critical shear stresses from sides to base: $\tau_{cs} = k \tau_{cb}$

We see that if $k < 0.75$ the sides will fail first.

if $\phi = 35^\circ$, $k < 0.75 \Rightarrow \alpha > 22.3^\circ$ or $m < 2.4$

So, for side slopes steeper than $m=2.4$, the sides will fail first.

Example: design of a stable channel.

A channel is to be lined with material having a representative size of 28mm. The cross section is to be trapezoidal and the bed slope is 0.0008. The angle of repose of the lining material is 35°. The maximum expected flow is 75 m³/s. Design a suitable cross section.

Solution:

For this problem you will need to use Shields diagram, Manning's formula, and Stricklers' equations.

1. Choose side slopes $\alpha < \phi$. For example, use 2H:1V or $m=2$,
so $\alpha = \tan^{-1}(1/2) = 26.6^\circ$

$$k = \left[1 - \frac{\sin^2 \alpha}{\sin^2 \phi} \right]^{1/2} = 0.626$$

2. Calculate k , so $k < 0.75$ hence **sides** are critical.
3. Calculate a depth y for the channel. This will depend on what critical shear stress you choose; here is an example:

a. Require $0.75\gamma y S \leq k[\tau_c]_{\text{bed}} = k[\gamma R S]_c$
applied $\leq$ resisting

b. Choose $d=11RS$ (for coarse material) – from previous section.

so, $0.75\gamma y S \leq k[\tau_c]_{\text{bed}} = k\gamma d/11$

therefore,

$$y \leq \frac{kd}{0.75 \cdot 11 \cdot S} = \frac{0.626 \cdot 0.028}{0.75 \cdot 11 \cdot 0.0008}$$

$$y \leq 2.66\text{m}$$

This is a maximum allowable value for y , so we should use a safety factor. Reduce this by 20% to give $y = 2.12\text{m}$

4. Determine the section base width (b) by using Manning's formula.

$$Q = \frac{1}{n} R^{2/3} A S^{1/2}$$

$$R = A/P$$

$$A = by + my^2$$

$$P = b + 2(m^2 + 1)^{1/2} y$$

$$\therefore Q = \frac{1}{n} \cdot \frac{(by + my^2)^{5/3}}{(b + 2 \cdot \sqrt{m^2 + 1} \cdot y)^{2/3}} \cdot S^{1/2}$$

From Strickler $n = 0.012 \cdot d_{75(mm)}^{1/6} = 0.0209$

$$\therefore 75 = \frac{1}{0.0209} \cdot \frac{(b \cdot 2.12 + 2 \cdot 2.12^2)^{5/3}}{(b + 2 \cdot \sqrt{2^2 + 1} \cdot 2.12)^{2/3}} \cdot 0.0008^{1/2}$$

$$\therefore 75 = 4.73 \cdot \frac{(b + 4.24)^{5/3}}{(b + 9.48)^{2/3}}$$

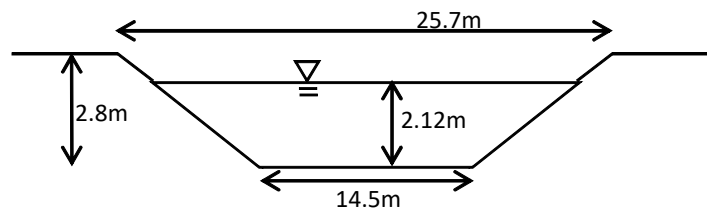
Use trial and error to solve for b. A first guess could be done by ignoring the 4.24 and 9.48 so that $b = 75/4.73 = 15.9\text{m}$

Trial and error:

b	15	14	14.5
Q	77.5	72.9	75.2

use $b = 14.5\text{ m}$

Need to allow for free board (waves or surges if a power channel) so should leave at least 0.5m free board.



Velocity (peak): $V = 75 / (14.5 \times 2.12 + 2 \times 2.12^2) = 1.89\text{m/s}$

Check shear stresses:

On bed: $\tau_{ob} = \gamma y S = 9810 \times 0.0008 \times 2.12 = 16.6\text{ N/m}^2$

Critical: $\tau_{cb} = 0.056(\gamma_s - \gamma)d = 0.56 \times 9810 \times (2.65 - 1) \times 0.028 = 25.4\text{ N/m}^2$

On sides: $\tau_{os} = 0.75 \gamma y S = 12.5\text{ N/m}^2$

Critical: $\tau_{cs} = k \tau_{cb} = 15.9\text{ N/m}^2$

Hence stable on both bed and sides!!

Entrainment using sediment size vs. shear stress.

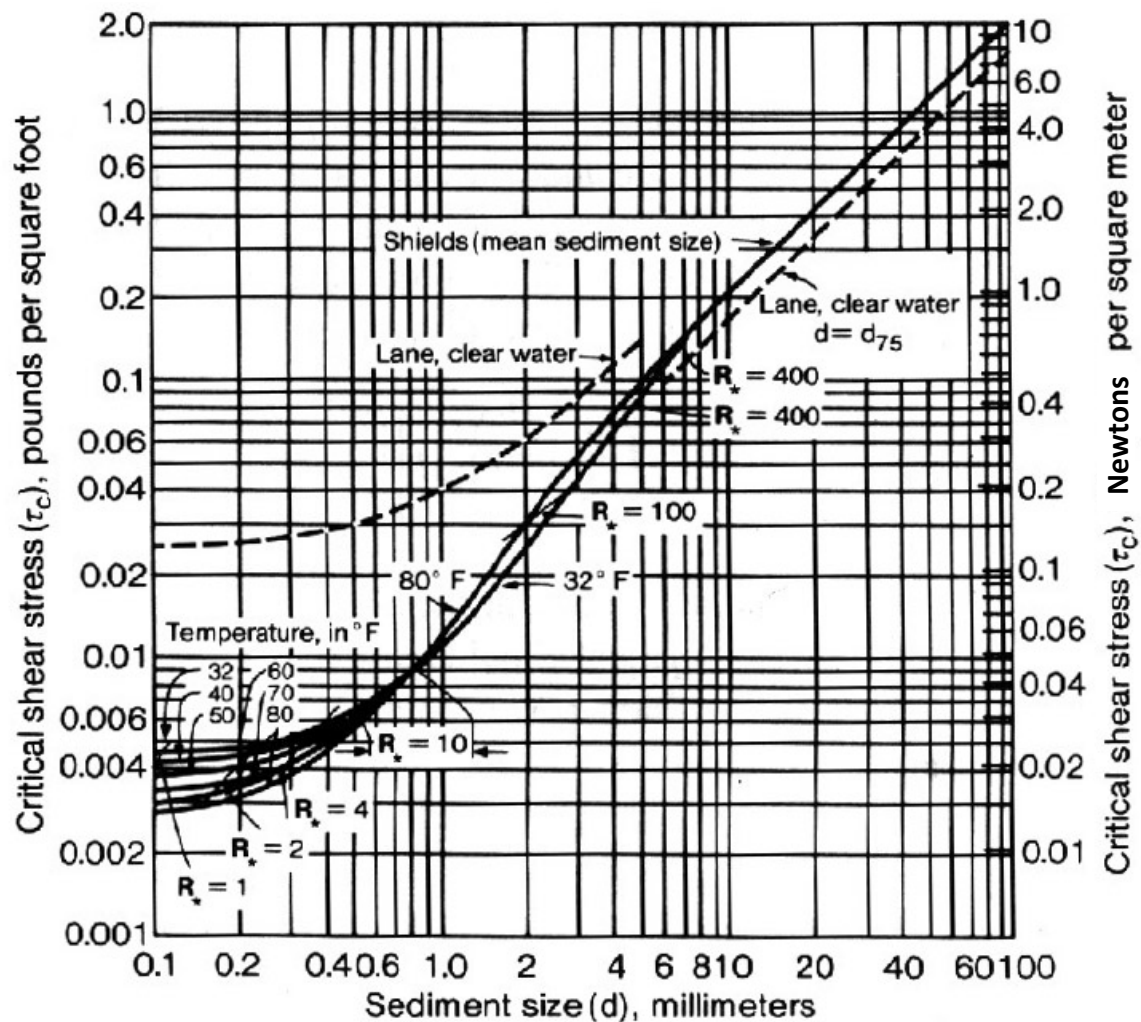


Figure 4. Critical shear stress for quartz sediment in water as function of grain size (Chang 1988) originally presented in the ASCE sedimentation manual.

References

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